Tejas Jain

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Education

Graphic Era Deemed to be University — Dehradun, Uttarakhand
B.Tech. in Computer Science Engineering; CGPA: 9.54/10 — *2025 – 2029*

DPSG Dehradun, Selaqui — Dehradun, Uttarakhand
Class 12 (CBSE), PCM – 96.20% — *2025*

Doon International School — Dehradun, Uttarakhand
Class 10 (CBSE) – 98.16% — *2023*

Technical Skills

Languages: Python, TypeScript/JavaScript, C, C++, SQL
Backend & APIs: FastAPI, Express.js (Node), Prisma ORM, REST, JWT/OAuth authentication, Webhooks
Frontend: React, Vite, Tailwind CSS, shadcn/ui
Data & AI: PostgreSQL, pgvector (RAG), Anthropic/OpenAI APIs, LLM agent pipelines
Computer Vision: OpenCV, YOLO, CNNs, Object Detection, Semantic Segmentation, Image Classification, PyTorch, TensorFlow/Keras
Infra & Tools: Docker, GitHub Actions (CI/CD), Nginx, Git, Linux

Projects

AI Revenue Recovery | *Python, FastAPI, React, PostgreSQL, pgvector* | GitHub — Razorpay AI Buildathon

- **Problem:** Merchants silently lose revenue to failed payments, abandoned checkouts, and overdue invoices, with no automated way to diagnose why or safely win it back.
- **Solution:** Built a 4-agent LLM pipeline (detect → diagnose → recover → audit) that classifies failures into 9 root causes via a pgvector RAG + LLM diagnosis engine, then runs bounded recovery – retry caps, human-approval gates, and a fraud-cluster detector that halts suspicious batches for review – with every decision logged to an audit trail.
- **Testing methods:** A 280+ test backend suite, plus a sandboxed Playground/Simulate mode – an AI customer persona conversing live with the recovery AI agent, and a manual mode where a human tester role-plays the customer against the same recovery AI agent – to validate agent behavior without touching real metrics.
- **Results:** Recovered ~19% (Rs. 37.8K of Rs. 198.6K) of at-risk revenue on a 76-event seeded batch, shipped with a GitHub Actions CI/CD pipeline building and pushing Docker images on every merge.

Gymodoro | *React, TypeScript, Express, Prisma, PostgreSQL* | gymodoro.com | GitHub — Personal Project

- People working long desk sessions take their breaks by scrolling their phone, staying sedentary and returning less focused than before. Built Gymodoro, a fitness + productivity app that turns every work break into a short movement window instead.
- **Features:** Pomodoro work/break timer, randomized break-time exercise suggestions with GIF/video/muscle-diagram guides, session analytics dashboard (focus time, calories burned, completion rate), JWT + Google OAuth authentication.
- **Tech stack:** React 19, TypeScript, Tailwind CSS, Express, Prisma, PostgreSQL, Docker Compose, Nginx, Oracle Cloud, Certbot/Let's Encrypt.

Self-Driving Car | *Python, OpenCV, YOLO, CNN, CARLA, Raspberry Pi* — Personal Project

CARLA Simulator | Project Video | GitHub

- **Problem:** Drive a car using only a camera – no LiDAR or radar – by fusing lane position, obstacles, and navigation cues into one real-time steering decision.
- **Solution:** Used a pretrained segmentation model to detect drivable area and OpenCV to extract waypoint arrows for navigation; engineered a multi-modal steering fusion algorithm computing weighted angular offsets from road-mask centroids, waypoints, and YOLO-detected obstacle boxes to drive steering and throttle.
- **Results:** Boosted inference 6–10× (3-4 to 20-40 FPS) by switching the detection backbone from SSD-MobileNet-v2 to YOLO, enabling smoother real-time autonomous driving in CARLA.

Raspberry Pi RC Car | Project Video | GitHub

- **Problem:** Take the simulated self-driving pipeline off-screen and onto a physical car, constrained by a Raspberry Pi's limited compute and real motor hardware.
- **Solution:** Collected driving data in a simulator, trained a Behavioral Cloning CNN on it, and deployed the model on a Raspberry Pi driving a differential-drive chassis (camera → CNN → steering angle → pigpio motor controller → per-wheel velocities); also built a servo-steering variant and a manual multi-key control mode (pynput) with incremental steering as a fallback.
- **Results:** Got both autonomous and manual driving modes working on real hardware, and picked up hands-on intuition for PWM tuning, motor control, and the sim-to-real gap.

Achievements

- Qualified JEE Advanced (2025); admission offers from IIT Mandi (Integrated MBA, Data Science & AI) and NIT